

Unifying Special Relativity and Quantum Mechanics

Lecture 4 — Spin, $SU(2)$ vs $SO(3)$, the Pauli–Lubanski Vector, and the Wigner Rotation

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–3: normal text and **yellow boxes** are a faithful transcription of the handwritten notes; **blue boxes** are my additions. This lecture continues the second page-numbering sequence started in Lecture 3, so the markers run [p. 9*] through [p. 32*].

Where we left off. Lecture 3 built, for the massive case, the rest-frame states $|M, \vec{0}, j, m\rangle$: eigenstates of P^μ (eigenvalue $p_{\text{ref}}^\mu = (M, \vec{0})$), of the Casimir P^2 (eigenvalue M^2), and of the little-group operators $|\vec{J}|^2, J^3$. It ended mid-sentence announcing a “less ambitious” intermediate goal — which is precisely the first line below. This lecture completes the entire construction, ending with the celebrated *Wigner rotation* formula for the action of a Lorentz transformation on one-particle states.

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1 Irreps of the angular momentum algebra (recap) [p. 9*]

... “less ambitious” goal, and try to understand how do they transform under a subgroup of Lorentz given by the little group.

Let us recall some angular momentum properties. We diagonalize $|\vec{J}|^2$ and J^3 by writing

$$\begin{aligned} |\vec{J}|^2 |j, m\rangle &= j(j+1) |j, m\rangle & J^2 &\equiv -i(J^+ - J^-)/2 \\ J^3 |j, m\rangle &= m |j, m\rangle & J^1 &\equiv (J^+ + J^-)/2 \\ J^\pm |j, m\rangle &= \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle \end{aligned}$$

We remark three points:

1) In group theory language,

$$V_j \equiv \left\{ |j, m\rangle, -j \leq m \leq +j \right\} \quad \text{with fixed } j$$

is the representation space corresponding to an irrep of the algebra.

2) The action of the generators on these states is given by the eigenvalue equations. The generators are given by $(2j+1)$ -dimensional matrices which [p. 10*] are given by (in components)

$$(J^k)_{m'm} \equiv \langle j, m' | J^k | j, m \rangle \quad (j \text{ fixed, } (2j+1)\text{-dim matrix}).$$

These matrix elements can be obtained from the eigenvalue equations.

Example $j = \frac{1}{2}$: two-dimensional representation.

$$J^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Example $j = 1$: three-dimensional representation.

$$J^3 = \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}, \quad J^1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J^2 = \frac{-i}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Added explanation (not in the original notes)

Where these matrices come from. Combine $J^1 = \frac{1}{2}(J^+ + J^-)$ and $J^2 = -\frac{i}{2}(J^+ - J^-)$ with the ladder formula: e.g. for $j = \frac{1}{2}$, $J^+ |\frac{1}{2}, -\frac{1}{2}\rangle = |\frac{1}{2}, +\frac{1}{2}\rangle$ (coefficient $\sqrt{\frac{3}{4} + \frac{1}{4}} = 1$) and $J^+ |\frac{1}{2}, +\frac{1}{2}\rangle = 0$, giving $(J^+)_{m'm} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; then $J^{1,2}$ follow, and one recognizes $J^k = \frac{1}{2}\sigma^k$ with σ^k the *Pauli matrices*. For $j = 1$ the ladder coefficients are $\sqrt{2}$, producing the $1/\sqrt{2}$ normalizations shown. (The $j = 1$ matrices are related to the antisymmetric generators $(J^k)_{ij} = -i\varepsilon_{kij}$ of Lecture 2 by a change of basis from Cartesian x, y, z to spherical components $m = +1, 0, -1$: same representation, different basis.) The ordering convention of the basis is $m = +j, \dots, -j$ from top to bottom.

3) The action of *group elements* on V_j can be obtained by exponentiating the generators:

$$D_j(\vec{\alpha}) = \exp(-i \vec{\alpha} \cdot \vec{J}), \quad [D_j(\vec{\alpha})]_{m'm} = \langle j, m' | \exp(-i \vec{\alpha} \cdot \vec{J}) | j, m \rangle.$$

Example: generic j , $\vec{\alpha} = \alpha \hat{z}$. [p. 11*]

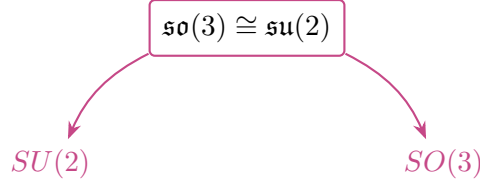
$$D_j(\alpha, \hat{z}) = \exp(-i \alpha \hat{z} \cdot \vec{J}) = \exp(-i \alpha J^3) \quad \text{with} \quad J^3 = \text{diag}(+j, +j-1, \dots, -j).$$

The exponentiation is trivial and gives

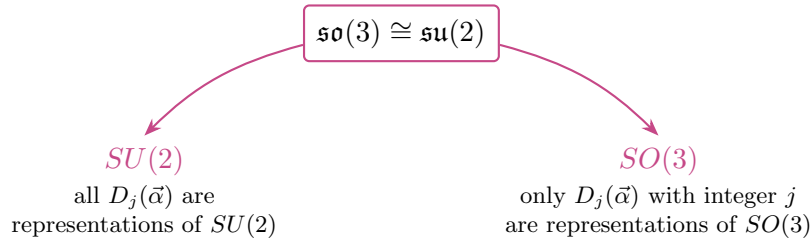
$$D_j(\alpha, \hat{z}) = \begin{pmatrix} e^{-i\alpha j} & & & \\ & e^{-i\alpha(j-1)} & & \\ & & \ddots & \\ & & & e^{+i\alpha j} \end{pmatrix}.$$

2 $SU(2)$ vs $SO(3)$: one algebra, two groups

There is an interesting conceptual point here. The goal is to obtain representations of the *group* by exponentiating representations of the *algebra*. However, in this case the same algebra is shared by two groups:



$SU(2)$ and $SO(3)$ are locally isomorphic (close to the origin) but, as manifolds, they [p. 12*] differ globally ($SU(2)$ is simply connected while $SO(3)$ is doubly connected). This fact has consequences from the point of view of representations:



Added explanation (not in the original notes)

The geometry behind “doubly connected”. As a manifold $SU(2)$ is the 3-sphere S^3 (a 2×2 unitary matrix with unit determinant is $a\mathbb{I} + i\vec{b} \cdot \vec{\sigma}$ with $a^2 + |\vec{b}|^2 = 1$): every loop can be shrunk to a point — simply connected. $SO(3)$ is S^3 with antipodal points identified (real projective space $\mathbb{RP}^3$), because the two-to-one map $SU(2) \rightarrow SO(3)$ sends $\pm U$ to the same rotation; a path from U to $-U$ in $SU(2)$ projects to a *closed* loop in $SO(3)$ that cannot be contracted (the famous “belt trick”), while going around it twice can — doubly connected. $SU(2)$ is the *universal (double) cover* of $SO(3)$, and its kernel $\{\mathbb{I}, -\mathbb{I}\}$ is exactly the $\pm$ ambiguity that half-integer representations pick up below.

How do we choose the values of j ? We said that the little group is $SO(3)$; does it mean that we should only take integer values of j ? $\hookrightarrow$ In reality, we cannot really say that the little group is $SO(3)$:

- 1) the theorem we proved works at the level of the *algebra*, and tells nothing about the global structure of the group;
- 2) [p. 13*] alternatively, we identified the little group as $SO(3)$ by looking at transformations that leave invariant $p_{\text{ref}}^\mu = (M, \vec{0})$. However, this is implicitly done in the representation in which rotations act on vectors, that is the representation with $j = 1$, shared between the two groups.

$\hookrightarrow$ These arguments cannot distinguish between the two groups.

To make a decision, we follow two distinct perspectives (that will eventually agree).

1) The “group theory perspective”: the group $SU(2)$ is “more fundamental” than $SO(3)$. It seems natural to consider $SU(2)$ and include all representations D_j .

2) The “QM perspective”: we consider $SO(3)$, and include only representations D_j with integer values of j .

What about the other “ j ”? Why do D_j with semi-integer values of “ j ” fail to be representations of $SO(3)$? [p. 14*]

Consider the following example. Take $D_j(\alpha, \hat{z})$ and perform two transformations:

$$D_j(\alpha_1, \hat{z}) D_j(\alpha_2, \hat{z}) = \begin{pmatrix} e^{-i(\alpha_1+\alpha_2)j} & & & \\ & e^{-i(\alpha_1+\alpha_2)(j-1)} & & \\ & & \ddots & \\ & & & e^{+i(\alpha_1+\alpha_2)j} \end{pmatrix} = D_j(\alpha_1 + \alpha_2, \hat{z}).$$

Consider the case in which $\alpha_1 + \alpha_2 = 2\pi$:

$$D_j(\alpha_1, \hat{z}) D_j(\alpha_2, \hat{z}) = (-1)^{2j} \mathbb{K}_{2j+1} = D_j(2\pi, \hat{z}).$$

We get

$$D_j(2\pi, \hat{z}) = (-1)^{2j} \mathbb{K}_{2j+1}.$$

2π -rotation around $\hat{z}$: if D_j is a group homomorphism, it should give the identity. *This is only true if j is integer.* For semi-integer j , we have

$$D_j(\alpha_1, \hat{z}) D_j(\alpha_2, \hat{z}) \Big|_{\alpha_1+\alpha_2=2\pi} = (e^{i\pi}) \mathbb{K}$$

$\hookrightarrow$ it “fails” to be a representation because of a *phase factor*. These are called “*projective representations*”.

[p. 15*] **Question: do we really need to exclude these representations?** In QM, physical states are defined by *rays* in the Hilbert space, that is up to a (global) phase.

$\rightsquigarrow$ In QM, the phase factor that makes D_j with semi-integer j a projective representation is *unphysical*.

$\rightsquigarrow$ We do not have a reason to exclude them. Both approaches, therefore, lead to the same conclusion: all values of $j = 0, \frac{1}{2}, 1, \dots$ are acceptable.

“Group theory perspective”: consider $SU(2)$ as little group, including all its representations ($j = 0, \frac{1}{2}, 1, \dots$)

$\equiv$ “QM perspective”: consider $SO(3)$ as little group but including both ordinary ($j = 0, 1, \dots$) and projective ($j = \frac{1}{2}, \frac{3}{2}, \dots$) representations.

Added explanation (not in the original notes)

This resolves the caveat raised in the Lecture 3 blue box about half-integer j . The deeper statement (Bargmann’s theorem, not covered in the notes): in quantum mechanics one should represent the *universal cover* of the symmetry group, because projective representations of a group are ordinary representations of its cover (when, as here, the algebra admits no nontrivial central extensions). For the Poincaré group the universal cover replaces $SO^\uparrow(1, 3)$ by $SL(2, \mathbb{C})$ and, inside it, the little group $SO(3)$ by $SU(2)$ — the two perspectives of the notes are two ways of saying this. The $(-1)^{2j}$ under a 2π rotation is not a formal curiosity: it is physically detectable in *interference* (neutron interferometry has measured the sign flip of a spin- $\frac{1}{2}$ state rotated by 2π), even though it is invisible on a single ray.

2.1 The eigensystem with spin

[p. 16*]

We thus write the eigensystem:

$$\begin{aligned}
P^\mu |M, \vec{0}, s, \sigma\rangle &= p_{\text{ref}}^\mu |M, \vec{0}, s, \sigma\rangle \\
P^2 |M, \vec{0}, s, \sigma\rangle &= M^2 |M, \vec{0}, s, \sigma\rangle \\
|\vec{J}|^2 |M, \vec{0}, s, \sigma\rangle &= s(s+1) |M, \vec{0}, s, \sigma\rangle & s = 0, \frac{1}{2}, 1, \dots \\
J^3 |M, \vec{0}, s, \sigma\rangle &= \sigma |M, \vec{0}, s, \sigma\rangle & -s \leq \sigma \leq +s
\end{aligned}$$

■ Formally, $|M, \vec{0}, s, \sigma\rangle = |M, \vec{0}\rangle \otimes |s, \sigma\rangle$, with P^μ and P^2 acting on $|M, \vec{0}\rangle$ and $|\vec{J}|^2$ and J^3 acting on $|s, \sigma\rangle$.

■ We relabel $(j, m) \rightarrow (s, \sigma)$. In the rest frame $\vec{P} = \vec{0}$ and $\vec{J}$ corresponds to spin.

The next step is the generalization of the above states with vanishing spatial momentum to states with generic $\vec{p} \neq \vec{0}$.

$\hookrightarrow$ This is not automatic, however, since for generic $\vec{p}$ we have that $\vec{P}$ and $\vec{J}$ do not commute.

$\hookrightarrow$ Before proceeding, therefore, we have to face this problem.

The key idea is the following. The Poincaré group admits a second Casimir operator.

3 The Pauli–Lubanski vector

[p. 17*]

$$\begin{aligned}
W^\mu &\equiv -\frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} J_{\nu\rho} P_\sigma & \text{Pauli–Lubanski vector,} & \varepsilon^{0123} = +1, \\
W^2 &= W_\mu W^\mu & \text{Casimir operator.}
\end{aligned}$$

Properties. ■ By direct computation, we find

$$[W^\mu, P^\nu] = 0.$$

We write

$$\begin{aligned}
[W^\mu, P^\nu] &= -\frac{1}{2} \varepsilon^{\mu\rho\lambda\sigma} [J_{\rho\lambda} P_\sigma, P^\nu] = -\frac{1}{2} \varepsilon^{\mu\rho\lambda\sigma} \left(J_{\rho\lambda} [P_\sigma, P^\nu] + [J_{\rho\lambda}, P^\nu] P_\sigma \right) \\
&= -\frac{1}{2} \varepsilon^{\mu\rho\lambda\sigma} g_{\rho\alpha} g_{\lambda\beta} [J^{\alpha\beta}, P^\nu] P_\sigma = -\frac{1}{2} \varepsilon^{\mu\rho\lambda\sigma} g_{\rho\alpha} g_{\lambda\beta} i (P^\alpha g^{\nu\beta} - P^\beta g^{\nu\alpha}) P_\sigma \\
&= -\frac{i}{2} \varepsilon^{\mu\rho\lambda\sigma} \left(\underbrace{P_\rho \delta_\lambda^\nu P_\sigma}_{\substack{\text{symmetric under the exchange} \\ \rho \leftrightarrow \sigma, \text{ while } \varepsilon^{\mu\rho\lambda\sigma} \text{ is antisymmetric}}} - \underbrace{P_\lambda \delta_\rho^\nu P_\sigma}_{\substack{\text{symmetric under exchange} \\ \lambda \leftrightarrow \sigma, \text{ while } \varepsilon^{\mu\rho\lambda\sigma} \text{ is antisymmetric}}} \right) = 0.
\end{aligned}$$

[p. 18*] ■ Consequently, we have $[P^\mu, W^\nu] = 0$. ■ Since W^2 is a scalar,

$$[W^2, J^{\mu\nu}] = 0.$$

Added explanation (not in the original notes)

Why “ W^2 is a scalar” settles it: W^μ is built covariantly out of tensors $(J_{\nu\rho}, P_\sigma, \varepsilon)$, so under conjugation it transforms as a four-vector, $D(\Lambda)^{-1} W^\mu D(\Lambda) = \Lambda^\mu{}_\nu W^\nu$ (this is also used explicitly on p. 24* below); hence the contraction $W_\mu W^\mu$ is invariant, which at the infinitesimal level is precisely $[W^2, J^{\mu\nu}] = 0$. Together with $[W^2, P^\mu] = 0$ (from $[W^\mu, P^\nu] =$

0), W^2 commutes with all ten generators: a genuine second Casimir, as promised. In the $[W^\mu, P^\nu]$ computation, note the recurring trick (used already on p. 17 of Lecture 1 and p. 30 of Lecture 2): a symmetric pair of indices contracted with the totally antisymmetric ε gives zero.

■ In components, the Pauli–Lubanski vector reads

$$W^0 = \vec{J} \cdot \vec{P}, \quad \vec{W} = \vec{J} P^0 + \vec{K} \times \vec{P}.$$

For instance, let us check the first one:

$$W^0 = -\frac{1}{2} \varepsilon^{0\lambda\mu\sigma} J_{\lambda\mu} P_\sigma = -\frac{1}{2} \varepsilon_{ijk} J_{ij} P_k = -\frac{1}{2} \varepsilon_{ijk} J^{ij} (-P^k) = \frac{1}{2} \varepsilon_{ijk} J^{ij} P^k = J^k P^k = \vec{J} \cdot \vec{P},$$

using $J_{ij} = J^{ij}$ (two spatial indices lowered, two sign flips) and $P_k = -P^k$, together with $J^k = \frac{1}{2} \varepsilon_{kij} J^{ij}$.

We make an important observation. If we apply W^μ on the state with reference momentum, we get

$$\begin{aligned} W^0 |M, \vec{0}, s, \sigma\rangle &= \vec{J} \cdot \vec{P} |M, \vec{0}, s, \sigma\rangle = 0, \\ \vec{W} |M, \vec{0}, s, \sigma\rangle &= (\vec{J} P^0 + \vec{K} \times \vec{P}) |M, \vec{0}, s, \sigma\rangle = M \vec{J} |M, \vec{0}, s, \sigma\rangle, \end{aligned}$$

from which it follows that [p. 19*]

$$W^3 |M, \vec{0}, s, \sigma\rangle = M\sigma |M, \vec{0}, s, \sigma\rangle,$$

$$W^2 |M, \vec{0}, s, \sigma\rangle = \left((\cancel{W^0})^2 - |\vec{W}|^2 \right) |M, \vec{0}, s, \sigma\rangle = -M^2 |\vec{J}|^2 |M, \vec{0}, s, \sigma\rangle = -M^2 s(s+1) |M, \vec{0}, s, \sigma\rangle.$$

Consequently we can use

$$\begin{aligned} W^2 |M, \vec{0}, s, \sigma\rangle &= -M^2 s(s+1) |M, \vec{0}, s, \sigma\rangle \quad \text{instead of the eq. for } |\vec{J}|^2, \\ W^3 |M, \vec{0}, s, \sigma\rangle &= M\sigma |M, \vec{0}, s, \sigma\rangle \quad (\star) \quad \text{instead of the eq. for } J^3. \end{aligned}$$

The twist is the following: W^2 is a Casimir and $[W^\mu, P^\nu] = 0 \rightsquigarrow$ we can use these operators also discussing the case $\vec{p} \neq \vec{0}$.

Let us just rewrite the equation for W^3 in a more covariant form. We introduce

$$S_{\hat{z}}^\mu \equiv (0, 0, 0, 1)$$

so that

$$W_\mu S_{\hat{z}}^\mu = W_0 \cancel{S_{\hat{z}}^0} + W_i S_{\hat{z}}^i = W_3 = -W^3,$$

and equation $(\star)$ becomes

$$\boxed{-\frac{W_\mu S_{\hat{z}}^\mu}{M} |M, \vec{0}, s, \sigma\rangle = \sigma |M, \vec{0}, s, \sigma\rangle}$$

[p. 20*] All in all, we have the eigensystem in the form

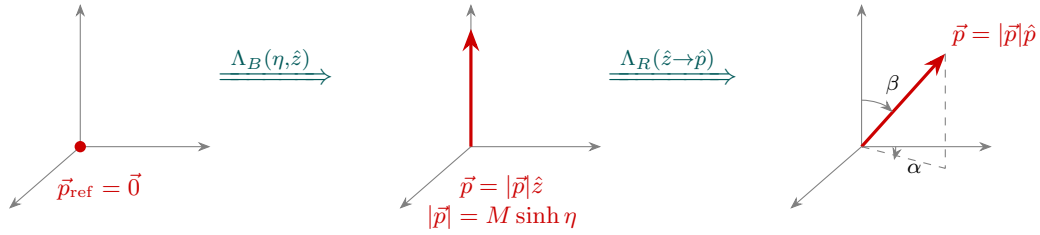
$$\begin{aligned}
P^\mu |M, \vec{0}, s, \sigma\rangle &= p_{\text{ref}}^\mu |M, \vec{0}, s, \sigma\rangle \\
P^2 |M, \vec{0}, s, \sigma\rangle &= M^2 |M, \vec{0}, s, \sigma\rangle \\
-\frac{S_{\hat{z}}^\mu W_\mu}{M} |M, \vec{0}, s, \sigma\rangle &= \sigma |M, \vec{0}, s, \sigma\rangle \\
W^2 |M, \vec{0}, s, \sigma\rangle &= -M^2 s(s+1) |M, \vec{0}, s, \sigma\rangle
\end{aligned}$$

Added explanation (not in the original notes)

The strategy in one sentence. The rest-frame labels (s, σ) were defined through $|\vec{J}|^2$ and J^3 , which do *not* commute with $\vec{P}$ and therefore cannot label moving states; the notes have now traded them for W^2 and $-S_{\hat{z}}^\mu W_\mu/M$, which act *identically* on the rest-frame states but, commuting with P^μ , remain good (simultaneously diagonalizable) operators for any momentum. The vector $S_{\hat{z}}^\mu$ is the rest-frame *spin quantization axis* promoted to a four-vector; note it satisfies $S \cdot S = -1$ and $S \cdot p_{\text{ref}} = 0$, properties preserved by the boost below. This confirms quantitatively the anticipation in the Lecture 3 blue box: $W^2 = -M^2 s(s+1)$ is “the covariant way of saying the spin is s ”.

4 Step 5: states with different momentum — the standard boost

We start from some considerations in momentum space:



(Angles: β = polar angle from $\hat{z}$, α = azimuthal angle of $\hat{p}$.)

First step:

$$p_{\text{ref}}^\mu \longrightarrow \Lambda_B(\eta, \hat{z}) p_{\text{ref}} = \begin{pmatrix} M \cosh \eta \\ 0 \\ 0 \\ M \sinh \eta \end{pmatrix}.$$

[p. 21*]

Second step:

$$\Lambda_B(\eta, \hat{z}) p_{\text{ref}} \longrightarrow \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z}) p_{\text{ref}} = \begin{pmatrix} M \cosh \eta \\ M \sinh \eta \hat{p} \end{pmatrix}.$$

So we define the transformation

$$\begin{aligned}
H(\hat{p}, |\vec{p}|/M) &\equiv \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z}) \quad (\text{depends on 3 parameters: } \eta \text{ and } \hat{p}), \\
H(\hat{p}, |\vec{p}|/M)^\mu{}_\nu p_{\text{ref}}^\nu &= p^\mu.
\end{aligned}$$

$\hookrightarrow$ Instead of η , we use $|\vec{p}|/M$ since they are related by $\sinh \eta = |\vec{p}|/M$.
 p is a generic momentum obtained from p_{ref} . It satisfies

$$p = \left((|\vec{p}|^2 + M^2)^{1/2}, |\vec{p}| \hat{p} \right).$$

Now there is an important conceptual point. H^μ_ν is a transformation given by a 4-by-4 matrix acting on 4-vectors. We now introduce the following operator: [p. 22*]

$$\mathcal{U}_H \equiv \mathcal{U}_H(\hat{p}, |\vec{p}|/M)$$

as the operator that corresponds to the action of the transformation $H(\hat{p}, |\vec{p}|/M)$ in the representation of the Lorentz group acting on states $|M, \vec{0}, s, \sigma\rangle$.

Added explanation (not in the original notes)

$H(\hat{p}, |\vec{p}|/M)$ is called the *standard boost* in the literature: a fixed, conventional choice of Lorentz transformation taking p_{ref} to each point p of the mass shell (here: pure boost along $\hat{z}$, then the rotation $\hat{z} \rightarrow \hat{p}$ constructed in Lecture 2, p. 41). Any other choice (e.g. a pure boost directly along $\hat{p}$) would work and would just redefine the basis states by a little-group rotation — the physics is convention-independent, but formulas like the Wigner rotation below refer to the chosen H . Note the exact parallel with $SE(2)$: there the “standard transformation” reaching a generic $\vec{p}$ on the circle was $\mathcal{U}(\theta)$; here it is $\mathcal{U}_H$, and the same conjugation trick with $D(\Lambda)^{-1} P^\mu D(\Lambda) = \Lambda^\mu_\nu P^\nu$ (Lecture 3, p. 50) is about to be used.

What is the action of this operator? What is $\mathcal{U}_H(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \sigma\rangle$?

► To understand this point, let us compute the action of P^μ on the previous state: In the following chain we abbreviate $\mathcal{U}_H \equiv \mathcal{U}_H(\hat{p}, |\vec{p}|/M)$ and $H \equiv H(\hat{p}, |\vec{p}|/M)$:

$$\begin{aligned} P^\mu \mathcal{U}_H |M, \vec{0}, s, \sigma\rangle &= \mathcal{U}_H \underbrace{\mathcal{U}_H^{-1} P^\mu \mathcal{U}_H}_{= H^\mu_\nu P^\nu} |M, \vec{0}, s, \sigma\rangle \\ &= \mathcal{U}_H H^\mu_\nu p_{\text{ref}}^\nu |M, \vec{0}, s, \sigma\rangle = p^\mu \mathcal{U}_H |M, \vec{0}, s, \sigma\rangle, \end{aligned}$$

where in the second line we used the eigenvalue equation $P^\nu |M, \vec{0}, s, \sigma\rangle = p_{\text{ref}}^\nu |M, \vec{0}, s, \sigma\rangle$ of the reference state. [p. 23*] The state $\mathcal{U}_H |M, \vec{0}, s, \sigma\rangle$ is still an eigenstate of P^μ , with transformed momentum $p^\mu = H^\mu_\nu p_{\text{ref}}^\nu$. We thus define

$$|M, \vec{p}, s, \sigma\rangle \equiv \mathcal{U}_H(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \sigma\rangle$$

(the action of $\mathcal{U}_H$ changes $\vec{0}$ into $\vec{p}$, leaving σ invariant). We can try to write an eigensystem for these states. We write immediately

$$\begin{aligned} P^\mu |M, \vec{p}, s, \sigma\rangle &= p^\mu |M, \vec{p}, s, \sigma\rangle \\ P^2 |M, \vec{p}, s, \sigma\rangle &= M^2 |M, \vec{p}, s, \sigma\rangle \\ W^2 |M, \vec{p}, s, \sigma\rangle &= -M^2 s(s+1) |M, \vec{p}, s, \sigma\rangle \end{aligned}$$

...but what about the last equation? In the moving frame, “ σ ” is no longer eigenvalue of the operator $-S_z^\mu W_\mu/M$. We need to find a “new” operator such that “ σ ” is its eigenvalue when acting on the states $|M, \vec{p}, s, \sigma\rangle$.

5 The spin operator for moving states; helicity

We start from a definition. Let us introduce [p. 24*]

$$S_p^\mu \equiv H^\mu_\nu S^\nu_{\hat{z}}; \quad S^\mu_{\hat{z}} = (H^{-1})^\mu_\nu S^\nu_p = H^\mu_\nu S^\nu_p$$

Then, consider the operator $-\frac{S_p^\mu W_\mu}{M}$:

$$-\frac{S_p^\mu W_\mu}{M} |M, \vec{p}, s, \sigma\rangle = -\frac{S_p^\mu W_\mu}{M} \mathcal{U}_H(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \sigma\rangle = -\frac{S_p^\mu}{M} \mathcal{U}_H(\mathcal{U}_H^{-1} W_\mu \mathcal{U}_H) |M, \vec{0}, s, \sigma\rangle$$

(W_μ is a four-vector, it transforms under Lorentz as P_μ does: $\mathcal{U}_H^{-1} W_\mu \mathcal{U}_H = H^\nu_\mu W_\nu = (H^{-1})^\nu_\mu W_\nu$)

$$\begin{aligned} &= -\frac{S_p^\mu}{M} \mathcal{U}_H (H^{-1})^\nu_\mu W_\nu |M, \vec{0}, s, \sigma\rangle = \mathcal{U}_H \left(\underbrace{-\frac{S_p^\mu (H^{-1})^\nu_\mu}{M}}_{= -\frac{S^\nu_{\hat{z}} W_\nu}{M}} \right) |M, \vec{0}, s, \sigma\rangle \\ &= \sigma \mathcal{U}_H |M, \vec{0}, s, \sigma\rangle = \sigma |M, \vec{p}, s, \sigma\rangle. \end{aligned}$$

The last equation that closes the eigensystem is

$$-\frac{S_p^\mu W_\mu}{M} |M, \vec{p}, s, \sigma\rangle = \sigma |M, \vec{p}, s, \sigma\rangle.$$

[p. 25*] **Question: what is the physical meaning of this equation?** To answer, let us compute explicitly $S_p^\mu W_\mu$.

■ First of all, we have $S_p^\mu = H^\mu_\nu S^\nu_{\hat{z}}$, $S_{\hat{z}} = (0, 0, 0, 1)$:

$$S_p = \Lambda_R(\hat{z} \rightarrow \hat{p}) \underbrace{\Lambda_B(\eta, \hat{z})}_{S_{\hat{z}}} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \cosh \eta & 0 & 0 & \sinh \eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \eta & 0 & 0 & \cosh \eta \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \sinh \eta \\ 0 \\ 0 \\ \cosh \eta \end{pmatrix}$$

With $\cosh \eta = p^0/M$, $\sinh \eta = |\vec{p}|/M$:

$$= \Lambda_R(\hat{z} \rightarrow \hat{p}) \begin{pmatrix} |\vec{p}|/M \\ p^0 \\ \frac{p^0}{M} \hat{z} \end{pmatrix} = \begin{pmatrix} |\vec{p}|/M \\ p^0 \\ \frac{p^0}{M} \hat{p} \end{pmatrix} = \begin{pmatrix} |\vec{p}|/M \\ p^0 \vec{p} \\ \frac{p^0}{M} |\vec{p}| \end{pmatrix} \rightsquigarrow \boxed{S_p^\mu = \left(\frac{|\vec{p}|}{M}, \frac{p^0 \vec{p}}{M |\vec{p}|} \right)}$$

■ We now compute $S_p^\mu W_\mu$ as follows:

$$-\frac{S_p^\mu W_\mu}{M} = -\frac{1}{M} (S_p^0 W_0 + S_p^i W_i) = -\frac{1}{M} (S_p^0 W^0 - S_p^i W^i) = -\frac{1}{M} (S_p^0 W^0 - \vec{S}_p \cdot \vec{W})$$

[p. 26*] (recall $W^0 = \vec{J} \cdot \vec{P}$, $\vec{W} = \vec{J} P^0 + \vec{K} \times \vec{P}$):

$$-\frac{S_p^\mu W_\mu}{M} = -\frac{1}{M^2} \left[|\vec{p}| \vec{J} \cdot \vec{P} - \frac{p^0 \vec{p}}{|\vec{p}|} \cdot (\vec{J} P^0 + \vec{K} \times \vec{P}) \right].$$

■ We apply this operator to an eigenstate $|M, \vec{p}, s, \sigma\rangle$:

$$\begin{aligned}
-\frac{S_p^\mu W_\mu}{M} |M, \vec{p}, s, \sigma\rangle &= -\frac{1}{M^2} \left[|\vec{p}| \vec{J} \cdot \vec{p} - \frac{p^0 \vec{p}}{|\vec{p}|} \cdot (\vec{J} p^0 + \vec{K} \times \vec{p}) \right] |M, \vec{p}, s, \sigma\rangle \\
&= -\frac{1}{M^2 |\vec{p}|} \underbrace{\left(|\vec{p}|^2 - (p^0)^2 \right)}_{= |\vec{p}|^2 - (|\vec{p}|^2 + M^2) = -M^2} \vec{J} \cdot \vec{p} |M, \vec{p}, s, \sigma\rangle \\
&= -\frac{1}{M^2 |\vec{p}|} (-M^2) \vec{J} \cdot \vec{p} |M, \vec{p}, s, \sigma\rangle = \frac{\vec{J} \cdot \vec{p}}{|\vec{p}|} |M, \vec{p}, s, \sigma\rangle.
\end{aligned}$$

$$\vec{J} \cdot \vec{P} |M, \vec{p}, s, \sigma\rangle = |\vec{p}| \sigma |M, \vec{p}, s, \sigma\rangle.$$

We thus find that σ represents the projection of the spin along the direction of motion: the helicity. We use “ λ ” instead of “ σ ” to denote the helicity.

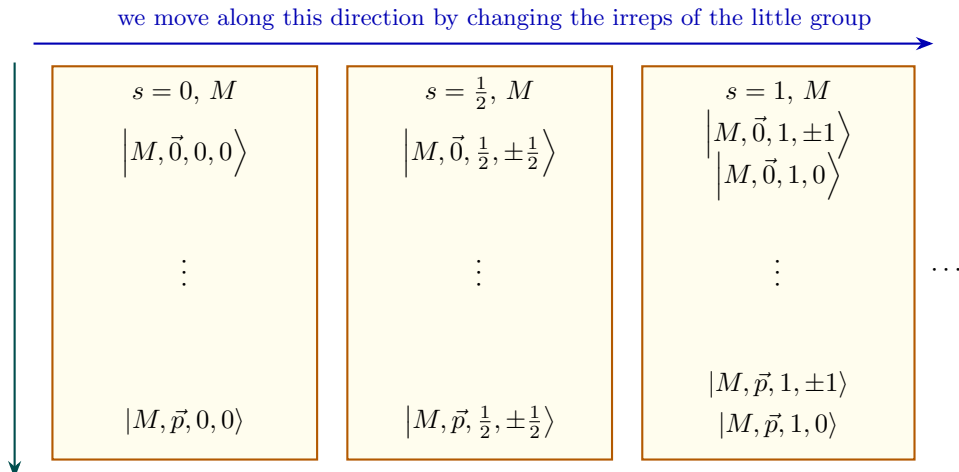
Added explanation (not in the original notes)

Reading the result. Dividing by $|\vec{p}|$: the operator $\vec{J} \cdot \hat{p}$ (component of the angular momentum along the momentum direction) has eigenvalue $\sigma \equiv \lambda$ on the moving states. This had to come out: the standard boost was “boost along $\hat{z}$, then rotate $\hat{z} \rightarrow \hat{p}$ ”, and the rest-frame label σ was the spin projection along $\hat{z}$ — the construction transports the quantization axis along with the momentum, so the label survives as the projection *along* $\hat{p}$. Two things worth knowing: (i) the cancellation $\vec{p} \cdot (\vec{K} \times \vec{p}) = 0$ is why the messy boost part of $\vec{W}$ drops out — helicity involves only $\vec{J}$; (ii) for *massive* particles helicity is a basis choice, not an invariant (an observer can overtake the particle and flip $\hat{p}$, hence λ); for massless particles (next topic of the course, presumably) it becomes a Lorentz invariant and is the only spin label. Also note S_p^μ obeys $S_p \cdot S_p = -1$, $S_p \cdot p = 0$: it is the standard *covariant spin vector*.

6 Step 6: unitary irreps of Poincaré — the Wigner rotation [p. 27*]

$$\begin{aligned}
P^\mu |M, \vec{p}, s, \lambda\rangle &= p^\mu |M, \vec{p}, s, \lambda\rangle \\
P^2 |M, \vec{p}, s, \lambda\rangle &= M^2 |M, \vec{p}, s, \lambda\rangle \\
W^2 |M, \vec{p}, s, \lambda\rangle &= -M^2 s(s+1) |M, \vec{p}, s, \lambda\rangle & s = 0, \frac{1}{2}, 1, \dots \\
\vec{J} \cdot \vec{P} |M, \vec{p}, s, \lambda\rangle &= \lambda |\vec{p}| |M, \vec{p}, s, \lambda\rangle & -s \leq \lambda \leq +s
\end{aligned}$$

Let us organize the above states as follows:



We move “vertically” generating states with generic momentum using Lorentz transformations that do not belong to the little group.

The claim is the following. If we consider the subspace spanned by the states [p. 28*]

$$\left\{ |M, \vec{p}, s, \lambda\rangle \right\}_{M,s \text{ fixed}}$$

it furnishes the representation space of a unitary irrep of the Poincaré group.

■ To check the validity of the claim, there are three important points:

- 1) if we consider a generic Poincaré transformation $\mathcal{U}(\Lambda, a)$ and act on the state $|M, \vec{p}, s, \lambda\rangle$, we should get a state with the same values of “ M ” and “ s ” (so to stay within the same subspace) while generating all possible values of $\vec{p}$ and λ

$$\rightsquigarrow \left\{ |M, \vec{p}, s, \lambda\rangle \right\}_{M,s} \text{ minimally invariant subspace under Poincaré;}$$

- 2) the action of $\mathcal{U}(\Lambda, a)$ must verify the composition rule of Poincaré;

- 3) after introducing an inner product, we must get that the action of $\mathcal{U}(\Lambda, a)$ is unitary.

[p. 29*] ■ It is crucial to understand the action of $\mathcal{U}(\Lambda, a)$ on the states $|M, \vec{p}, s, \lambda\rangle$. Using the factorization $\mathcal{U}(\Lambda, a) = \mathcal{U}(a)\mathcal{U}(\Lambda)$, we study separately translations and Lorentz transformations.

6.1 Action of space-time translations

$$\mathcal{U}(a) = \exp(i a_\mu P^\mu), \quad \mathcal{U}(a) |M, \vec{p}, s, \lambda\rangle = e^{i a_\mu P^\mu} |M, \vec{p}, s, \lambda\rangle = e^{i a_\mu p^\mu} |M, \vec{p}, s, \lambda\rangle.$$

$$\mathcal{U}(a) |M, \vec{p}, s, \lambda\rangle = e^{i a \cdot p} |M, \vec{p}, s, \lambda\rangle, \quad \mathcal{U}(a+b) = \mathcal{U}(a)\mathcal{U}(b)$$

► we stay within the same subspace; the composition rule is verified.

6.2 Action of Lorentz transformations

Goal: understand what is $\mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle$. Let us act on this state with P^μ :

$$\begin{aligned} P^\mu \mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle &= \mathcal{U}(\Lambda) \mathcal{U}(\Lambda)^{-1} P^\mu \mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle \\ &= \mathcal{U}(\Lambda) \Lambda^\mu_\nu P^\nu |M, \vec{p}, s, \lambda\rangle = \mathcal{U}(\Lambda) \Lambda^\mu_\nu p^\nu |M, \vec{p}, s, \lambda\rangle. \end{aligned}$$

[p. 30*] Introducing $\boxed{p_\Lambda^\mu \equiv \Lambda^\mu_\nu p^\nu}$:

$$= \mathcal{U}(\Lambda) p_\Lambda^\mu |M, \vec{p}, s, \lambda\rangle = p_\Lambda^\mu \mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle.$$

$$\mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle \text{ is eigenstate of } P^\mu \text{ with eigenvalue } p_\Lambda \equiv \Lambda p.$$

However, we notice that, based on our definition (p. 23*), we know already how to write an eigenstate of P^μ with eigenvalue p_Λ : it is given by

$$|M, \vec{p}_\Lambda, s, \lambda\rangle = \mathcal{U}_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/M) |M, \vec{0}, s, \lambda\rangle.$$

Consequently, there must be a relation between $\mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle$ and $|M, \vec{p}_\Lambda, s, \lambda\rangle$. Let’s find it out:

$$\begin{aligned} \mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle &= \mathcal{U}(\Lambda) \mathcal{U}_H(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \lambda\rangle \\ &= \mathcal{U}_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/M) \underbrace{\left[\mathcal{U}_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/M)^{-1} \mathcal{U}(\Lambda) \mathcal{U}_H(\hat{p}, |\vec{p}|/M) \right]}_{\text{we focus on this chain of transformations}} |M, \vec{0}, s, \lambda\rangle \stackrel{(*)}{=} \end{aligned}$$

Let us consider what happens in momentum space: [p. 31*]

$$p_{\text{ref}} \xrightarrow{H} p = H p_{\text{ref}} \xrightarrow{\Lambda} p_{\Lambda} = \Lambda p \xrightarrow{H_{\Lambda}^{-1}} p_{\text{ref}}$$

The combined transformation $H(\hat{p}_{\Lambda}, |\vec{p}_{\Lambda}|/M)^{-1} \Lambda H(\hat{p}, |\vec{p}|/M)$ leaves p_{ref} invariant $\rightsquigarrow$ it must be an element of the little group!

$$H(\hat{p}_{\Lambda}, |\vec{p}_{\Lambda}|/M)^{-1} \Lambda H(\hat{p}, |\vec{p}|/M) \equiv W(\Lambda, p) \in SO(3) \quad \underline{\text{Wigner rotation}}$$

(4×4 matrix acting on momentum-state four-vectors).

The operation

$$\mathcal{U}_H(\hat{p}_{\Lambda}, |\vec{p}_{\Lambda}|/M)^{-1} \mathcal{U}(\Lambda) \mathcal{U}_H(\hat{p}, |\vec{p}|/M)$$

will represent the action of the Wigner rotation on the reference states $|M, \vec{0}, s, \lambda\rangle$.

This is a turning point since we know how the states $|M, \vec{0}, s, \lambda\rangle$ transform under rotations. We write in general:

$$e^{-i\theta\hat{n}\cdot\vec{J}} |M, \vec{0}, s, \lambda\rangle = e^{-i\theta\hat{n}\cdot\vec{J}} |M, \vec{0}\rangle \otimes |s, \lambda\rangle = |M, \vec{0}\rangle \otimes e^{-i\theta\hat{n}\cdot\vec{J}} |s, \lambda\rangle =$$

[p. 32*]

$$\begin{aligned} &= |M, \vec{0}\rangle \otimes \sum_{\lambda'} |s, \lambda'\rangle \underbrace{\langle s, \lambda' | e^{-i\theta\hat{n}\cdot\vec{J}} |s, \lambda\rangle}_{=[D_s(\theta, \hat{n})]_{\lambda'\lambda}} \\ &= |M, \vec{0}\rangle \otimes \sum_{\lambda'} [D_s(\theta, \hat{n})]_{\lambda'\lambda} |s, \lambda'\rangle = \sum_{\lambda'} [D_s(\theta, \hat{n})]_{\lambda'\lambda} |M, \vec{0}, s, \lambda'\rangle. \end{aligned}$$

We go back to $\circledast$:

$$\begin{aligned} &\stackrel{\circledast}{=} \mathcal{U}_H(\hat{p}_{\Lambda}, |\vec{p}_{\Lambda}|/M) \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} |M, \vec{0}, s, \lambda'\rangle \\ &= \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} \underbrace{\mathcal{U}_H(\hat{p}_{\Lambda}, |\vec{p}_{\Lambda}|/M) |M, \vec{0}, s, \lambda'\rangle}_{=|M, \vec{p}_{\Lambda}, s, \lambda'\rangle}. \end{aligned}$$

We thus get

$$\mathcal{U}(\Lambda) |M, \vec{p}, s, \lambda\rangle = \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} |M, \vec{p}_{\Lambda}, s, \lambda'\rangle$$

► if we apply a Lorentz transformation, we do change $\vec{p}$ and λ but we stay within the same subspace.

Added explanation (not in the original notes)

This is the payoff of the entire course so far — Wigner’s formula. Read it in words: a Lorentz transformation acting on a one-particle state does two things, (i) it kicks the momentum, $\vec{p} \rightarrow \vec{p}_{\Lambda}$, and (ii) it rotates the spin index λ by the spin- s representation matrix of the *Wigner rotation* $W(\Lambda, p)$ — the little-group element measuring the mismatch between “boost, then transform” and “transform, then boost”. A few remarks:

- Structurally this is the exact Poincaré upgrade of the $SE(2)$ computation of Lecture 2

(p. 24): there the sandwich $\mathcal{U}(\phi + \theta)^{-1}\mathcal{U}(\phi)\mathcal{U}(\theta)$ was the *identity* (trivial little group, abelian $SO(2)$), so no index rotated; here the little group is non-abelian $SU(2)$ and the sandwich is a genuine rotation acting on the $2s + 1$ spin components. Comparing the two side by side is the clearest way to see what “induced representation” means.

- If Λ is itself a rotation R , one can check $W(R, p) = R$: rotations act on the spin index in the naive way. For Λ a boost, $W(\Lambda, p)$ is a nontrivial rotation depending on p — the *Wigner/Thomas rotation*, the same kinematic effect behind Thomas precession, rooted in $[K, K] \sim -iJ$ (Lecture 2, p. 38).
- The three checks of p. 28* are now in reach: (1) M, s unchanged and all $(\vec{p}_\Lambda, \lambda')$ reachable — minimal invariant subspace, i.e. irreducibility; (2) the composition rule follows from the “cocycle” property $W(\Lambda_2\Lambda_1, p) = W(\Lambda_2, \Lambda_1 p) W(\Lambda_1, p)$, which is a one-line consequence of the definition of W ; (3) unitarity follows as in Lecture 2 (p. 26) once the inner product $\langle M, \vec{p}', s, \lambda' | M, \vec{p}, s, \lambda \rangle \propto \delta_{\lambda'\lambda} \delta^3(\vec{p}' - \vec{p})$ is introduced, using that D_s is unitary — presumably the business of the next lecture.
- Bottom line of Wigner’s classification, now fully realized in the massive case: a relativistic particle *is* the pair (M, s) , and its state space is spanned by $|M, \vec{p}, s, \lambda\rangle$ with the transformation law in the yellow box.

(End of Lecture 4 — the construction of the massive unitary irreps is complete; the notes are expected to continue with the inner product/unitarity and the massless case.)